

# Hedonic price functions and the measurement of preferences: The case of Swedish wine consumers

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## Abstract

A hedonic price function for wine is estimated using Swedish data, 1989–91. Implicit prices for quality attributes are determined *not* from a regression of variety price on a vector of quality attributes, but rather from a regression of quantity sold (adjusted for weeks of availability) on price and quality attributes. Such a ‘reduced form’ is justified by the assumption that prices and attribute contents can be taken as exogenous to the Swedish consumer. Price elasticity is estimated to be about  $-1.65$  holding quality constant, showing that Swedish consumers are highly sensitive to price. Estimates of the implicit valuations of quality attributes are shown to differ greatly from those obtained from the more usual hedonic regression with price as the dependent variable.

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## 1. Introduction

Hedonic price analysis has its origins in agricultural economics. Frederick V. Waugh (1928) published his pioneering paper on quality factors influencing vegetable prices. (See also Waugh, 1929.) Waugh regressed the price per lot of asparagus in Boston (May–July, 1927) on three different dimensions of quality: measures of color, size of stalks, and uniformity of spears. His purpose was to determine consumers' relative valuations of these characteristics, which he regarded as useful information for asparagus producers. Later Court (1939) and Griliches (1961) ran similar regressions with the purpose of discovering consumers' preferences with respect to various options on the automobiles they purchased, so that an appropriate quality-adjusted measure of automobile price changes over time could be constructed. Chow (1967) and Cole et al. (1986) and Berndt and Griliches (1990) investigated the relation of computer prices to quality characteristics by similar methods and with a similar purpose.

In recent years, hedonic methods have been employed to measure consumer valuations of air quality and other environmental amenities by virtue of the fact that they are part of a bundle of various characteristics that a household purchases when buying a house. Regressions of housing prices on characteristics including these amenities, or disamenities, are then assumed to yield gradients which estimate the buyer's marginal willingness to pay for each attribute in much the same sense that Waugh interpreted his asparagus regressions. (See the survey of environmental economics by Cropper and Oates (1992, especially pp. 706–708); and the papers by Witte et al. (1979), Palmquist (1984), Bartik (1987), Epple (1987), Ohsfeldt (1988), Ohsfeldt and Smith (1985), and Kanemoto (1988).) The hedonic analysis of house prices by focusing on housing market equilibrium within a city, assuming an inelastic supply of housing, ignores the role of migration among areas in the determination of implicit valuations. Other studies, such as those of Smith (1983), Cropper and Arriaga-Salinas (1980) and Blomquist et al. (1988), take a longer-run view which assumes that workers can move freely from one location to another and estimate the wage differentials associated with different bundles of amenities associated with different locations.

The basic formulation of all of these analyses is to obtain observations on prices of varieties of a differentiated commodity, units of which embody varying amounts of different attributes or qualities, be these environmental amenities, greenness of stalk, or speed of multiplication. A regression, possibly nonlinear, is then used to estimate the so-called hedonic price function, the gradients of which are the implicit prices of the attributes, the ratios of which in turn are supposed to reflect consumers' marginal rates of substitution among attributes. Muellbauer (1974) presents several models of utility maximizing behavior which justify such interpretation of the hedonic price function in terms of underlying consumer preferences, but he is quick to admit that his analysis neglects the other, producer, side of the market. Just as in the case of ordinary demand analysis, using data on

prices and quantities, there is an unresolved identification problem in attempting to draw inferences about consumer preferences from hedonic price functions. As Working (1927) pointed out long ago, price–quantity observations represent equilibria of demand and supply and only under special circumstances can regressions of price on quantity or of quantity on price be used to infer anything about the underlying supply and/or demand functions. The classic paper by Rosen (1974) shows that essentially the same identification problem exists for hedonic price functions: Observations on variety prices and on the quantities of attributes associated with each variety and the number of units sold are, in general, jointly determined by demand and supply. The demand for attributes in various combinations is determined by consumer preferences, but the supplies of these attributes depend on the costs of producing them and on producers' access to the markets for varieties and to the scarce resources used in production. A large and statistically significant coefficient for a particular quality attribute in an estimated hedonic price function may reflect, not consumers' high valuation of that attribute, but rather the difficulties or high costs which producers have in achieving that attribute per se or in relation to other attributes. In general, the coefficients in the estimated hedonic regression reflect both supply and demand considerations, both producers' costs and consumers' preferences.

Just as in Working's analysis, it may be possible to identify the price–quantity relation with either demand or supply, so in the case of hedonic analysis it may be possible to identify the hedonic price function exclusively with factors reflecting consumer preferences or, alternatively, solely with factors reflecting costs of production and supply conditions. If the quantity supplied shifts exogenously independently of the shifts in the demand function, a regression of price on quantity will estimate the price elasticity of demand (Schultz, 1938). Conversely, if prices are exogenously determined, for example, by world prices determined in world markets, for a small group of consumers purchasing in a local market with supplies perfectly elastic at world prices, the regression of quantity on price may arguably be used to estimate the demand elasticity (Stone, 1954).

In the case of environmental amenities, automobile options, or computer characteristics it is often argued that supplies are exogenously determined. (Although recent developments in the environmental literature seek to achieve identification by other assumptions; see for example, Ohsfeldt (1988), Ohsfeldt and Smith (1985), Smith et al. (1990).) On the other hand, I know of no attempt to date to exploit situations in which consumers face exogenously determined prices in order to isolate consumers' implicit valuations of different quality attributes, i.e., of the hedonic analogue to Stone's studies of consumer demand in the United Kingdom. In this paper, I use a body of data for Sweden to estimate Swedish consumers' preferences for different attributes of wine, arguing that variety prices are determined by world supply and demand considerations and may be treated exogenously to the Swedish consumers of wine.

## 2. A general framework for hedonic analysis and the fundamental identification problem

Suppose a variety of some commodity is composed of  $n$  attributes,  $Z = Z_1, \dots, Z_n$ . Associated with the bundle of attributes, which define the variety, is a unit price  $P(Z)$ . There is no risk of confusion if I identify varieties and their prices by the vector index  $Z$  and also define an index of the ‘quality’ of each variety by a function or functions with the values of the elements of  $Z$  as arguments. In this section, I first look at the way in which the demand for a closely related group of varieties, such as different types of wine, can be derived from a simple utility maximizing model. This model is then combined with a parallel formulation for producers to reveal the identification problem as noted by Rosen (1974). A solution to the identification problem when consumers purchase many units of each variety at exogenously given prices is proposed and applied to Swedish data in the next section.

The model of consumer behavior does not restrict each to purchase only one variety or one unit of a particular variety, as in the Lancaster (1966) attribute model, and it does permit substitution among varieties. The formulation adopted here is based on what Muellbauer (1972, 1974) calls the variable ‘repackaging’ theory of quality differences proposed by Fisher and Shell (1971). Let  $Q(Z)$  be a vector of the varieties of a group of closely related commodities purchased by the representative consumer, and let  $X$  be a vector of purchases of other commodities.  $Z$  is supposed to vary continuously so that  $Q(Z)$  represents a ‘spectrum’ of varieties. Following Fisher and Shell, I suppose that associated with each point in the spectrum  $Q(Z)$  is an ‘index’ of quality, the elements in the vector  $A(Z)$ , also a spectrum, such that the consumer’s total utility is a separable function of the category utility function of the aggregate  $[A(Z), Q(Z)]$  and the quantities of other goods consumed.<sup>1</sup> The advantage of this formulation is that consumers can substitute continuously across varieties, with the result aggregated into a single number by the category utility function  $V$ . Formally the utility function can be written

$$U[V[A(Z), Q(Z)], X]. \quad (1)$$

The consumer maximizes  $U$  subject to an overall budget constraint, given the prices of the varieties indexed by  $Z$  and the prices of the other goods included in

<sup>1</sup>  $[A(Z), Q(Z)]$  means a function, not necessarily linear, of the elements of  $A(Z)$  and  $Q(Z)$ . The problem is that, together with linearity of the budget constraint, linearity of  $[A(Z), Q(Z)]$  implies that an optimizing consumer will choose only one variety and not a combination, except in the case where the elements of  $A(Z)$  are proportional to the elements of  $P(Z)$ . (See Muellbauer (1974, p. 989); ‘variable repackaging’, however, allows for continuous behavior; Muellbauer (1972). This was pointed out to me by F. Bourguignon.)

Of course, aggregate behavior may be continuous, even when individual behavior is not.

the vector  $X$ . If the utility function is also homothetic in  $V$  versus other goods, the prices of the other goods will not influence choices among varieties. These choices will be influenced only by relative variety prices. If the elements of  $A(Z)$  are supposed to be a common function of the elements of  $Z$ , say  $a(Z)$ , then, as Muellbauer shows, at the consumer optimum, the elements of  $P(Z)$  are proportional to the function  $a(Z)$ , where the constant of proportionality is the value of  $a(Z)$  times the ratio of the marginal utility of the composite  $V$  to the marginal utility of income. If the function  $U$  is not homothetic, the function  $a(Z)$  will generally vary with the individual's total utility or level of income and other characteristics. Consumers' valuations of various quality characteristics are expressed by the derivatives of  $a(Z)$  and not by the derivatives of  $P(Z)$  directly. Consumers take the function  $P(Z)$  as given and maximize utility (1) by their choice of the quantities of each variety they buy at those prices,  $Q(Z)$ . Thus, utility maximization yields a spectrum of demand:

$$Q(Z) = \varphi[P(Z), Y | a(Z)], \quad (2)$$

where  $Y$  is a vector of consumer characteristics such as income, age and other unobservables.

In a competitive market, the entire function  $P(Z)$  is determined as an equilibrium of the demand for, and supply of all of the various varieties, as Rosen shows, as are the quantities of the different varieties exchanged in the market,  $Q(Z)$ . Following this line of reasoning, we can derive a system of supply equations for the different varieties of the group of closely related varieties, which we can also define by a vector of quality indexes times the quantities produced, exactly in parallel to the derivation of (2). If the elements of the vector can be represented by a common function of the vector  $Z$ , say  $b(Z)$ , we arrive finally at the supply functions

$$Q(Z) = \psi[P(Z), W | b(Z)], \quad (3)$$

where  $W$  is a vector of producer characteristics reflecting access to scarce resources and other unobservable factors affecting the costs of producing scarce varieties. On the supply side  $b(Z)$  represents the difficulties of achieving the levels of quality defining the variety indexed by  $Z$ . In equilibrium, both  $Q(Z)$  and  $P(Z)$  are functions of consumer preferences for varieties, expressed in the index  $a(Z)$ , the costs of producing different varieties, expressed in the index  $b(Z)$ , and the vectors of consumer and producer characteristics,  $Y$  and  $W$ :

$$Q(Z) = F[a(Z), b(Z), Y, W], \quad (4)$$

$$P(Z) = G[a(Z), b(Z), Y, W]. \quad (5)$$

Usual practice in hedonic analysis is to estimate

$$P(Z) = G^*(Z, Y) \quad (6)$$

and to identify the gradient of  $G^*$  with the gradient of  $a(Z)$  reflecting consumer valuations of the quality attributes  $Z$ . Even if  $W$  were included, however, the function  $G^*(Z, Y, W)$  reflects *both*  $a(Z)$  and  $b(Z)$ .  $G^*$  is what is called a *confluent relationship* (Haavelmo, 1944, p. 29). The problem is exactly the same identification problem to which Working pointed: Inference on consumers' relative preferences for various attributes cannot, in general, be based on the regression of unit variety prices on the quantities of attributes per unit and other variables.<sup>2</sup>

If the quantities of varieties embodying different quantities of attributes are exogenously determined, so that market equilibrium is the result of (2) alone, solved for  $P(Z)$  as a function of  $Q(Z)$  and  $Y$ ,

$$P(Z) = \varphi^*(Z, Y), \quad (7)$$

then the gradient of  $\varphi^*$  does indeed represent consumer preferences alone, i.e., are proportional to the derivatives of  $a(Z)$  by a constant of proportionality which does not depend on other prices or income in the homothetic separable case. But even in this case, it is not clear why the number of units of each variety available is suppressed and unit price regressed on unit quantities of quality attributes,  $Z$ , and on  $Y$ , to obtain an estimate of  $\varphi^*$ .

In the general case, in which  $P(Z)$  and  $Q(Z)$  are jointly determined by (4) and (5), Rosen suggests estimating  $P(Z)$  as a function of  $Z$  by the usual hedonic method, without regard to  $Y$  and  $W$ , then differentiating it to obtain estimates of the equilibrium implicit prices, then estimating the structural systems (2) and (3) in a second stage. Apart from the fact that data on  $Y$  and  $W$  associated with different varieties are frequently not obtainable, Rosen's procedure is flawed. As Brown and Rosen (1982) argue (p. 766), the implicit price estimates cannot play the same role in estimation that direct observations on the implicit prices would play if available: "Because such constructed prices are created only from observed sample quantities, [so that] any *new* information that they may provide... can only come from *a priori* restrictions placed on the functional form of the price function  $P(Z)$ . In the absence of such additional restrictions, the second-stage 'structural' estimation of the sort suggested by Rosen may only reproduce the information already provided by the first-stage estimation of the  $P(Z)$  function". Brown and Rosen go on to provide an example of a quadratic hedonic price function which, if estimated by OLS, leads to estimated implicit prices, which, in turn, if used in the way Rosen suggests, exactly reproduce the parameters already obtained by estimating  $P(Z)$ .

On the other hand, in the polar case in which variety prices are exogenously determined and variety supplies are perfectly elastic to the group of consumers

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<sup>2</sup> Another problem may be the homotheticity assumptions necessary to justify considering one price for the varieties in the group indexed by  $Z$ .

being considered, properties of consumers preferences, i.e., the demand functions  $\varphi$ , can be inferred from regressions of the quantities of each variety consumed on its prices and associated unit quality attributes and any other variables which characterize consumers of each variety. Eq. (3) is irrelevant in this case since the variety prices are exogenous to the group of consumers considered. Thus, in this polar case, we should regress the quantities of each variety consumed on the unit variety price and the measures of quality attributes which characterize that variety,

$$Q(Z) = R[P(Z), a(Z), Y]. \quad (8)$$

Consumer preferences,  $a(Z)$ , are expressed by the effects of the unit quality attributes of the quantities of each variety purchased, holding unit variety price constant. But the coefficients of the unit quality attributes in (8) are not directly comparable with the implicit prices derived by differentiating  $G^*$  above, i.e., the slopes of the quality attributes in the usual hedonic function. To obtain a comparable result, we have to ask how much extra a consumer would be willing to pay for a little more of attribute  $i$  holding the quantities of all varieties fixed: That result is obtained from the implicit function theorem as

$$\pi_i = -(\partial R / \partial Z_i) / (\partial R / \partial P). \quad (9)$$

The values of  $\pi_i$  reflect consumers' relative preferences for the different quality attributes, *given variety prices*.

In what follows, I argue that the data for Swedish consumers of wine, which I have been able to obtain, fits the second polar case, in which variety prices are exogenous and consumer preferences are expressed by the quantities of each variety which they buy, and I show that the implicit valuations of consumers estimated from regressions based on (8),  $\pi_i$ , differ greatly from those (incorrectly) estimated by the usual hedonic method from (6). Since many situations are approximated by this polar case, the method proposed here is of general significance and may be applicable in numerous situations in which the quantities of each variety are not exogenously determined.

### 3. A case study of Swedish wine consumers

Wine is a highly differentiated product. Descriptions of its characteristics and the way in which it is produced are available, for example, in Amerine and Roessler (1983) and Johnson (1991). It is sold and purchased at prices which vary over an extremely wide range, and, arguably, produced and consumed under reasonably competitive conditions viewed world-wide. While data are readily available on prices and, unsystematically, on quality variations, such data are rarely linked. Except for data problems, wine would seem to be an ideal commodity for the study of price determination for a highly differentiated product under competitive conditions. Swedish consumers are such a small part of the

world market that world prices can effectively be treated as exogenous for them so that the appropriate way in which to assess their preferences for different quality characteristics of wine is from a regression of quantity on price and attribute measures rather than from the usual hedonic analysis which treats price as dependent. Doing so yields results which differ greatly as between the two approaches.

### *3.1. Description of the Swedish data*

In Sweden all wines and alcoholic beverages are centrally controlled and marketed through Vin och Sprit (V&S) and state stores, Systembolaget, at retail. No wine is domestically produced, V&S is the sole importer. Many wines are imported in bulk and bottled in Sweden; others are imported directly already bottled. There is a relatively simple markup formula for the retail price based on the F.O.B. price Stockholm, which includes a markup for taxes and marketing costs. Once the retail price is determined, it is not changed as inventories are depleted and stock-outs are frequent. With the exception of two wines which retail for over Skr. 1000 per bottle the prices of the wines in the sample here considered (adjusted to a standard 750 ml bottle; Skr. 6 approximately equals US\$1) range from a low of about Skr. 30 to a high of just under Skr. 600, with the majority clustering in the Skr. 40–60 range. These wines do not, therefore, represent the high end of the scale, that is those wines which are sold at auction and/or held for investment purposes. It seems inappropriate to treat them as a durable commodity either in the hands of Swedish consumers or of V&S; they are imported and purchased to be drunk.

About 7000 wines world-wide are screened by V&S field representatives each year, of which 5% or 350 are marketed in Sweden. To the extent that the 350 marketed is not a random sample of wines screened, there exists a selectivity bias in the results reported below for the effects of various measures of quality. If, for example, V&S representatives correctly perceive Swedish tastes, the effects of the quality measures will be underestimated; in the extreme case there would be no discernable effects of variation in what is measured. Unfortunately, no information whatsoever is available for wines rejected, so that no assessment of the potential selectivity bias is possible. For this analysis, I simply assumed the sample is random, bearing in mind that many variables may appear to be insignificant which in fact are significant.

Those wines which are chosen for marketing in Sweden, and only those wines, are subjected to careful chemical analyses and tasters' evaluations by a panel of nine tasters before offered through Systembolaget. The panel of tasters has remained the same through the sample period. They rate the wines independently on 17 separate taste evaluations categories of which 11 are available for this analysis. Data are in the form of scores averages over the nine tasters and are not



available for individual tasters. One category is an overall assessment of quality, independent of individual categories. This variable is imperfectly correlated with the other scores and several other variables including price which is significant although unknown to the tasters at the time of evaluation.<sup>3</sup> Presumably the variable captures various factors omitted in the evaluation process as well as nonlinearities in the relationship between these and prices or quantities purchased.

There are seven chemical analyses reported. Because of rounding in the data supplied, density almost never differs from 1.0 (the density of water) and has therefore been eliminated in the statistical analyses reported below. Sugar content is frequently not measured. Therefore, I have analyzed all data for which price, taste tests and chemical analyses are complete, omitting the sugar variable (239 observations), and the subsample for which a sugar measurement is recorded (199 observations) separately. The results are very similar and I report only those for the smaller sample for which the sugar measurement is present. In order to assess the independence of the information recorded in the chemical analyses and taste analyses, I ran a canonical correlation analysis between the two sets of measurements and found that the relationship between the two sets of variates was largely accounted for by the high, but not perfect correlation between tasters perceptions of sweetness and measured sugar content or precipitate. This analysis is omitted here.

V&S have generously made available data on nearly all the wines marketed during a 102-week period in 1989–91, a total of around 247 usable observations on prices, chemical analyses, and other variables described below. Of these, 106 are imported in bottles, the remainder in bulk and bottled in Sweden, 139 are red, 193 originate in the EEC countries, only 1 in the rest of Europe, 48 in the USA, Oceania, or Latin America, and the rest in unspecified areas. Quantities are also recorded in both liters and weeks of availability, ranging from lows of under 75 liters and one week to highs of nearly a million liters and a full 102 weeks. There are frequent stock-outs and there is a weak positive association between liters sold and weeks available, which suggests that many wines are in fact rationed by availability. Wines imported in bulk and bottled in Sweden are less subject to

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<sup>3</sup> The regression of tasters' overall evaluation on log price, and all the remaining chemical and taste analyses yields the following results: Price is not the most significant variable, but it is nonetheless highly significant and positively associated with the taster's overall evaluations. Other significant variables are whether the wine is imported in bulk (–), or in bottles (+), aroma (+), bouquet (+), sweetness (–), mouth feel (–), color (+), and typicality (+). Curiously, origin influences the tasters' overall evaluation only for the regression in which a sugar measurement is included. Sugar itself is not a significant factor when included, but both ethylacetate and extract are significant only for the larger sample when sugar is ignored. The  $R^2$ 's are 0.61 and 0.63, respectively. Thus, the overall evaluation may be included in regressions including other tasters' evaluations and prices because it clearly measures something different, at least in part.

stock-out.<sup>4</sup> Restriction to varieties for which quantity data are available further reduces sample size when these data are used.

Although it is often alleged by some that V&S exercises some monopsony power in some wine producing regions, there is no evidence of this. Thus, the F.O.B. price, and therefore the Swedish retail price, can be taken as exogenous to Swedish consumers. One further point has to do with whether V&S purchases in quantities which reflect their perceptions of Swedish tastes. This is arguably the case, so that stock-outs occur less frequently than would otherwise be the case, but this, as we shall see is not likely to affect the results significantly over and above the effects of stock-outs in general.

Thus, the *quantities* of wines with different characteristics are determined by the tastes of Swedish consumers and the constraints they face, taking *prices* as given. If V&S could, or did, obtain as much of any particular wine as consumers would demand at world price plus mark-up, the analysis of the previous section would suggest treating the quantity consumed of each wine as a function of its price and a vector of characteristics. Indeed, the number of liters sold in Sweden is available for part of the data set (235 observations). However, V&S is not always able to purchase all of every type of wine that Swedish consumers would buy at the prices charged. Many wines sell out after only a few weeks and only a few are sold for the full 102 weeks covered by the data. In principle, therefore, the theory of demand for a *rationed* commodity should be employed in the analysis. Although data on the distribution of sales through time from date of introduction in the state stores exist, they are not available to me. Thus, I cannot carry out an analysis along the lines suggested by Neary and Roberts (1981). Wines are in perfectly elastic supply up to stock-out and then simply unavailable.<sup>5</sup>

To approximate the number of liters which would be sold in 102 weeks if supplies were perfectly elastic at the price set by V&S, I compute

$$ADLIT = \text{liters sold} \times 102 / \text{weeks available}.$$

<sup>4</sup> As defined in Table 2, *WEEKS* indicates number of weeks available during the sample period, *B* is a dummy variable = 1, if the wine is imported in bulk, = 0 otherwise. A regression of *WEEKS* on *B* yields (199 observations)

$$WEEKS = 26.6 + 20.9B, \quad R^2 = 0.11.$$

(8.2)      (5.0)

Figures beneath the coefficients are *t*-ratios. Thus, wines which are imported in bulk are available on the average twice as long as those imported in bottles.

<sup>5</sup> To the extent a given wine runs out in some stores but not in others, we might want to view the additional search costs incurred by a consumer as reflecting a higher variety real price. Thus, the supply would not have a corner but rather an upward sloping segment between the perfectly elastic and the perfectly inelastic segments. Unfortunately, I have no information on the manner in which stock-outs actually occur except the fact that the pecuniary price of a given variety charged the consumer is never changed from date of inception.

Obviously, a wine in short supply, and only informally rationed, will generally sell out faster the shorter its supply. Thus *ADLIT* will be an *overestimate* of the number of liters which would have been sold. How is this error likely to be correlated with observed quality attributes and observed prices? In a complex multivariate situation, this is not easy to surmise. To venture a guess, I would say wines at the upper end of the price range are more likely to be subject to more severe rationing in Sweden. This would also be true for intrinsically higher quality wines, but, as attributes related to quality are measured with greater error, these effects are likely to be smaller. On balance, therefore, I would expect price elasticities of quantities demanded to be *biased downwards* in absolute value: Higher prices would be associated with spuriously high *adjusted* liters, i.e., not as low quantities as they should be. (Indeed, average prices are higher for wines which sell out quickly than for those available for longer periods.) As indicated above, wine imported in bulk is less subject to stock-out, so that the coefficient of a dummy = 1 for wine imported in bulk, = 0 otherwise, will be positively correlated with the error in measuring quantity; a positive coefficient would not indicate that Swedish consumers prefer wine imported in bulk but would simply reflect the fact that such wines are less subject to stock-out.

Both *TASTE* and *CHEMICAL* variables are listed in Table 1 by V & S mnemonic, together with the Swedish term, an English translation and a brief description. Table 2 lists other available variables used in the analyses reported below.

### 3.2. *Standard hedonic regressions versus regressions with adjusted quantities as dependent*

The usual hedonic regression in which unit variety prices are regressed on unit quality characteristics can be carried out using the Swedish data, but the coefficients in such regressions reflect a mixture of world consumers preferences and the world supply considerations, i.e., the costs and difficulties of achieving wines with the specified characteristics. Of course, to the extent that wines imported into Sweden are not a random sample of wines produced world-wide, there is a selectivity bias present even in the estimates of such confluent relationships.

In the case of a small group of consumers whose tastes differ from those of the market at large and who exert no appreciable influence on prices, however, inferences about their preferences can be based on analyses which treat variety prices as explanatory, quantities as dependent, and include measures of quality attributes. Casual observation suggests that the preferences of Swedish consumers differ greatly from those of the representative world consumer and from those of V & S tasters but it is difficult to prove this assertion rigorously. I assume that it is so, and show that inferences about Swedish wine consumers' preferences drawn from the usual hedonic regressions (Analysis A1: price transformed by taking logs, and Analysis A2: price transformed by a Box–Cox transform with parameter  $\lambda = -0.78$ ) differ greatly from those drawn from the alternative regressions of

Table 1  
Mnemonics for chemical and taste analyses

| Mnemonic                              | Swedish term   | English term       | Description                                                                                                |
|---------------------------------------|----------------|--------------------|------------------------------------------------------------------------------------------------------------|
| <i>Taste analysis</i> <sup>a</sup>    |                |                    |                                                                                                            |
| <i>FN</i>                             | Fargnyans      | Color              | In red wines ranging from red brown to blue. In white wines ranging from straw yellow to dark amber.       |
| <i>FI</i>                             | Färgintensitet | Color intensity    | Depth of color, lightness or luminance. Enhanced by clarity.                                               |
| <i>A</i>                              | Arom           | Aroma              | Odors in the wine originating in the grape.                                                                |
| <i>BO</i>                             | Bouquet        | Bouquet            | Odors in the wine originating in the fermentation process, particularly those that develop after bottling. |
| <i>SO</i>                             | Sötma          | Sweetness          | Dry: 0.15–0.4%; Demi-sec: 0.8–1.2%; Semi-sweet: 1.6–3.2%; Sweet: > 3.2%.                                   |
| <i>SY</i>                             | Syrlighet      | Acidity            | Taste of all volatile and nonvolatile acids; the opposite is "flat".                                       |
| <i>ST</i>                             | Strävhet       | Astringency        | Puckery tactile sensation. "Harsh" and "tannic" are related concepts, ≠ "bitter".                          |
| <i>K</i>                              | Kolsyra        | Mouthfeel          | Related to carbon dioxide content. Associated with characteristic odor of carbonic maceration.             |
| <i>F</i>                              | Fyllighet      | Body               | Tactile sensation differentiating low ethanol from high ethanol wines; "thin" vs. "full-bodied".           |
| <i>T</i>                              | Typiskhet      | Typicality         | Does it taste like it should? E.g., a merlot, a pinot noir, Bordeaux, Burgundy?                            |
| <i>BE</i>                             | Betyg          | Overall evaluation | Contribution and balance of all factors.                                                                   |
| <i>Chemical analyses</i> <sup>b</sup> |                |                    |                                                                                                            |
| <i>SOCKER</i>                         | Socketerhalt   | Sugar              | See SO. glucose, fructose, glycerol.                                                                       |
| <i>ALKH</i>                           | Alkoholhalt    | Alcoholic content  | Ethanol. Primary source of viscosity as well as inebriation.                                               |
| <i>DENS</i>                           | Denistet       | Density            | Relative to pure water = 1.0.                                                                              |
| <i>FLSYRA</i>                         | Flyktiga Syror | Volatile acidity   | Principally the more volatile acids, e.g., acetic and lactic acids.                                        |
| <i>ETAC</i>                           | Etylacetat     | Ethylacetate       | Source of vinegary odor. Also acetic acid. Objectionable in excess.                                        |
| <i>EXTR</i>                           | Extraktalt     | Extract            | Solids, mainly residual sugar not fermented out.                                                           |
| <i>TOTSYR</i>                         | Totalsyra      | Total acidity      | Lactic, formic, acetic and other aliphatic acids.                                                          |

<sup>a</sup> Taste variables are measured by each of nine tasters on a 10-point scale, 1 being the worst evaluation, 10 being the best; the data used in the analysis are the mean scores over all tasters for each wine.

<sup>b</sup> Variables are measured in appropriate units, usually parts per 1000 ml.

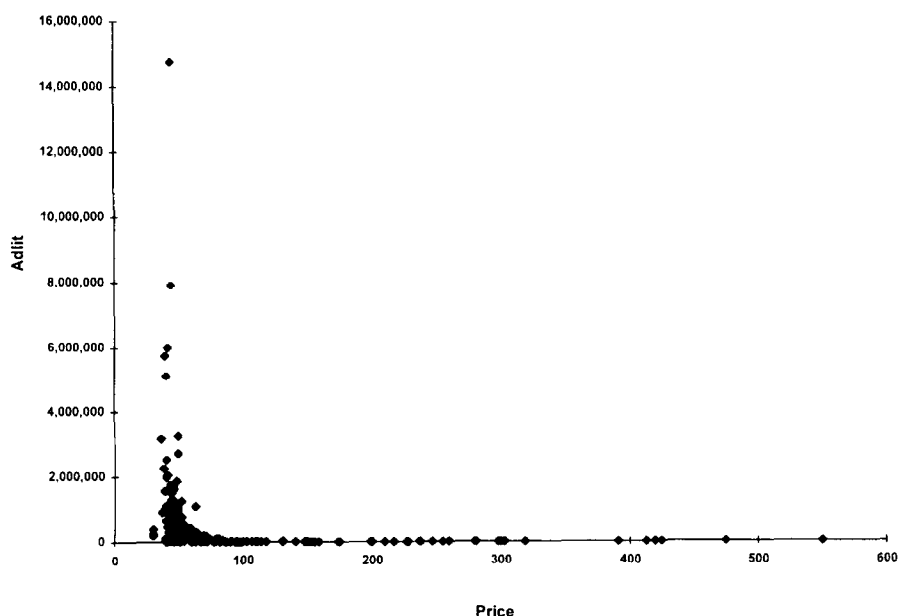
Table 2

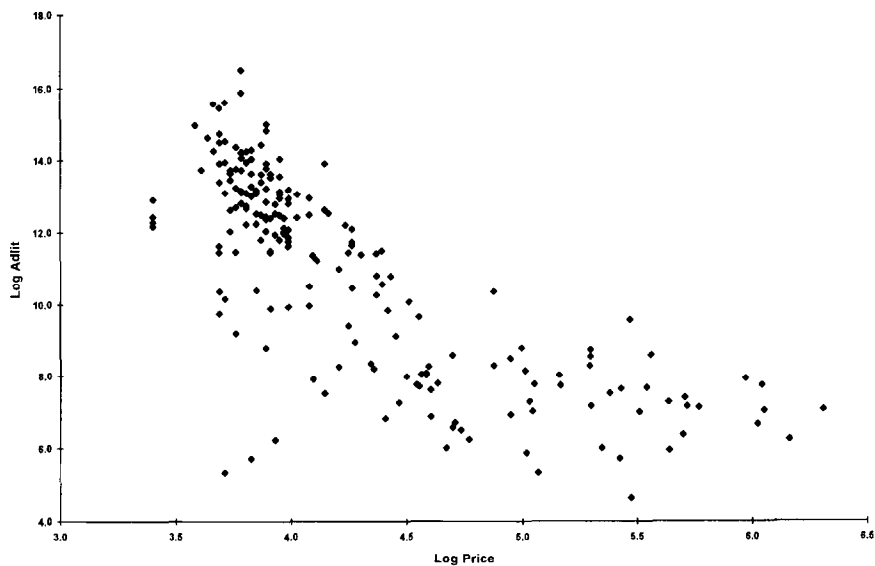
Other variables used

|                   |                                                                                       |
|-------------------|---------------------------------------------------------------------------------------|
| <i>PRICE (P)</i>  | Price in Skr. adjusted to a standard 750 ml. bottle. Not adjusted for bottle deposit. |
| <i>LITERS (Q)</i> | Liters sold over the 102 week period.                                                 |
| <i>WEEKS (W)</i>  | Weeks available for sale.                                                             |
| <i>ADLIT (Q')</i> | Adjusted liters = $Q \times (102 / W)$ .                                              |
| <i>RED (R)</i>    | 1 for red wines, 0 for white wines.                                                   |
| <i>OR1</i>        | = 1, for EEC origin; = 0, otherwise.                                                  |
| <i>OR2</i>        | = 1 for USA, Latin America, Australia and New Zealand origin; = 0, otherwise.         |
| <i>B</i>          | 1 if wine is imported in bulk, 0 if in bottles.                                       |

quantity on price and attribute measures (Analysis B1: double log in price and quantity, and Analysis B2: quantity transformed by a Box–Cox transform with  $\theta = 0.06$ , price by a B-C transform with  $\lambda = -0.78$ , values are obtained by maximum likelihood and both significantly different from 0, the double log case). These conclusions support my contention that the usual hedonic analysis leads to erroneous inferences about preferences.

The wines in my sample are concentrated in the 40–60 Skr. range but have a long-tailed distribution. Plots of adjusted liters sold, *ADLIT*, against price suggest

Fig. 1. Plot of *ADLIT* against *PRICE*

Fig. 2. Plot of log *ADLIT* against log *PRICE*

a highly curvilinear relation which may be better approximated by a double-log function than a linear one. Fig. 1 reports a plot of *ADLIT* against price; Fig. 2 reports the double log plot. There is still some curvilinearity present. Fitting

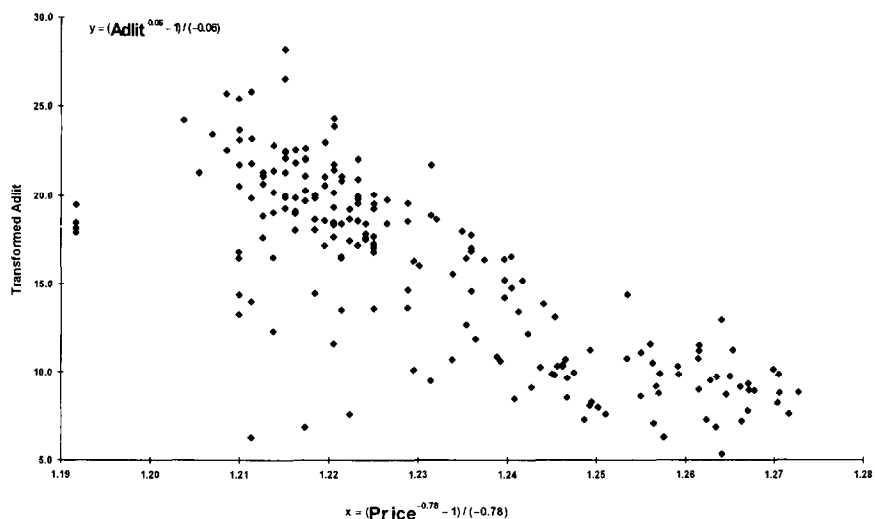
Fig. 3. Plot of transformed *ADLIT* against transformed *PRICE*

Table 3

Analysis A: OLS regressions of log  $P$  on tasters' evaluations, chemical analyses, and dummies (t-ratios in parentheses)

| Item                                         | Sample 235    | Subsample<br>(sugar included) 195 |
|----------------------------------------------|---------------|-----------------------------------|
| $R^2$                                        | 0.741         | 0.755                             |
| Residual $SS/df$                             | 26.115/215    | 22.781/174                        |
| <i>Coefficients of independent variables</i> |               |                                   |
| Intercept                                    | 2.664 (3.48)  | 2.502 (2.88)                      |
| $B$                                          | -0.450 (6.33) | -0.370 (4.51)                     |
| $R$                                          | -0.254 (1.48) | -0.619 (2.66)                     |
| $OR1$                                        | 0.259 (1.24)  | 0.355 (1.26)                      |
| $OR2$                                        | -0.063 (0.29) | -0.003 (0.01)                     |
| $ALKH$                                       | -0.058 (1.36) | -0.068 (1.36)                     |
| $FLSYRA$                                     | -0.023 (1.65) | -0.026 (1.53)                     |
| $ETAC$                                       | 0.007 (5.41)  | 0.007 (4.91)                      |
| $EXTR$                                       | 0.001 (0.34)  | 0.008 (1.17)                      |
| $TOTSYR$                                     | 0.0002 (0.12) | -0.001 (0.52)                     |
| $SOCKER$                                     | —             | -0.009 (1.26)                     |
| $FN$                                         | 0.003 (0.96)  | 0.005 (1.27)                      |
| $FI$                                         | 0.0007 (0.19) | -0.001 (0.30)                     |
| $A$                                          | -0.003 (0.87) | -0.004 (0.79)                     |
| $BO$                                         | 0.009 (2.05)  | 0.010 (2.09)                      |
| $SO$                                         | 0.0009 (0.16) | 0.003 (0.46)                      |
| $SY$                                         | -0.007 (1.42) | -0.006 (1.12)                     |
| $ST$                                         | 0.008 (1.89)  | 0.017 (2.97)                      |
| $K$                                          | 0.058 (1.97)  | 0.067 (2.17)                      |
| $F$                                          | 0.006 (0.08)  | -0.0007 (0.09)                    |
| $T$                                          | 0.008 (1.61)  | 0.009 (1.48)                      |
| $BE$                                         | 0.014 (2.38)  | 0.013 (1.93)                      |

models in which both price and quantity were transformed by a Box–Cox transform also yielded estimates of the B-C parameters of 0.06 for quantity and -0.78 for price, both significantly different from zero, which would be the double log formulation. A plot of transformed  $ADLIT$  against transformed price is presented in Fig. 3 and reveals very little curvilinearity. Table 3 presents the results of the usual hedonic analyses for the subsample in which  $SOCKER$  is excluded for Analysis A1 in which log $P$  is the dependent variable and Analysis A2 in which  $P$  is transformed to  $(P^{-0.78} - 1)/-0.78$ . In addition to the quality characteristics, I introduced dummies for wines originating in the EEC and for those originating in the USA, Latin America, Australia, or New Zealand, if the wine was imported in bulk or in bottles, and whether red or white.

The  $R^2$  are quite respectable. Alternative regressions based on a larger sample for which  $SOCKER$  was excluded gave similar results. The highlights are as follows: There is a significant price premium for wines imported in bottles over

wines imported in bulk. Red wines sell at a discount against white. The quantity of ethylacetate (*ETAC*) found is the most significant variable positively associated with price. While quite significantly positively associated with price, tasters' overall quality evaluation is much less significant than either the ethylacetate content or whether the wine is imported in bottles. Bouquet (*BO*), astringency (*ST*), and mouth feel (*K*) (largely a question of CO<sub>2</sub> content) are all significantly positively associated with price, volatile acidity' (*FLYSYRA*) is on the margin of significance and negatively associated with price. Neither origin dummy is significant by itself, however, the *F*-statistic is 12.85 for 2 and 174 degrees of freedom and is highly significant showing that the Swedish Consumer pays a premium for wines from the EEC and obtains New World wines at a discount. Other variables are individually insignificantly associated with log price. These relationships do

Table 4

Analysis B: OLS Regressions of  $\log Q^*$  on  $\log P$ , tasters' evaluations, chemical analyses and dummies (*t*-ratios in parentheses; implied implicit prices at mean *P*)

| Item                                         | Sample 198  |        | Implied normalized<br>implicit price <sup>a</sup> | Implied subsample<br>162 |        | Normalized<br>implicit<br>price <sup>a</sup> |
|----------------------------------------------|-------------|--------|---------------------------------------------------|--------------------------|--------|----------------------------------------------|
| <i>R</i> <sup>2</sup>                        | 0.729       |        | -                                                 | 0.814                    |        | -                                            |
| Residual <i>ss</i> / <i>df</i>               | 431.886/177 |        | -                                                 | 258.146/140              |        | -                                            |
| <i>Coefficients of independent variables</i> |             |        |                                                   |                          |        |                                              |
| Intercept                                    | 20.301      | (5.12) | -                                                 | 20.162                   | (5.41) | -                                            |
| Log <i>P</i>                                 | -1.675      | (4.80) | -1.0                                              | -1.665                   | (5.09) | -1.0                                         |
| <i>B</i>                                     | 1.984       | (5.11) | 1.184                                             | 2.116                    | (5.82) | 1.271                                        |
| <i>R</i>                                     | -1.365      | (1.67) | -0.815                                            | 0.068                    | (0.07) | 0.041                                        |
| <i>OR1</i>                                   | -1.540      | (1.63) | -0.919                                            | -0.407                   | (0.38) | -0.244                                       |
| <i>OR2</i>                                   | 0.143       | (0.15) | 0.085                                             | 0.934                    | (0.87) | 0.561                                        |
| <i>ALKH</i>                                  | 0.008       | (0.03) | 0.005                                             | -0.187                   | (0.87) | -0.112                                       |
| <i>FLSYRA</i>                                | 0.058       | (0.81) | 0.035                                             | 0.102                    | (1.44) | 0.061                                        |
| <i>ETAC</i>                                  | -0.0007     | (0.11) | -0.000                                            | -0.002                   | (0.34) | -0.001                                       |
| <i>EXTR</i>                                  | 0.019       | (1.11) | 0.011                                             | 0.083                    | (3.14) | 0.050                                        |
| <i>TOTSYR</i>                                | 0.031       | (3.29) | 0.019                                             | -0.037                   | (3.86) | -0.022                                       |
| <i>SOCKER</i>                                | -           |        | -                                                 | -0.093                   | (3.47) | -0.056                                       |
| <i>FN</i>                                    | -0.26       | (1.54) | -0.016                                            | -0.030                   | (1.84) | -0.018                                       |
| <i>FI</i>                                    | -0.044      | (2.53) | -0.026                                            | -0.025                   | (1.50) | -0.015                                       |
| <i>A</i>                                     | 0.014       | (0.69) | 0.008                                             | 0.009                    | (0.46) | 0.005                                        |
| <i>BO</i>                                    | -0.006      | (0.29) | -0.004                                            | -0.016                   | (0.78) | -0.010                                       |
| <i>SO</i>                                    | -0.048      | (1.78) | -0.029                                            | -0.011                   | (0.44) | -0.007                                       |
| <i>SY</i>                                    | 0.041       | (1.68) | 0.024                                             | 0.041                    | (1.73) | 0.025                                        |
| <i>ST</i>                                    | 0.038       | (1.82) | 0.023                                             | -0.007                   | (0.32) | -0.004                                       |
| <i>K</i>                                     | 0.068       | (0.46) | 0.041                                             | 0.109                    | (0.82) | 0.065                                        |
| <i>F</i>                                     | -0.0004     | (0.01) | -0.000                                            | -0.019                   | (0.54) | -0.011                                       |
| <i>T</i>                                     | -0.030      | (1.14) | -0.018                                            | -0.028                   | (1.08) | -0.017                                       |
| <i>BE</i>                                    | 0.032       | (1.10) | 0.019                                             | 0.049                    | (1.80) | 0.029                                        |

<sup>a</sup> Coefficient of quality variable divided by price elasticity of demand. See the text for an explanation.



not allow for interaction among the different quality characteristics. Unfortunately, attempts to introduce interaction effects did not yield insignificant results perhaps due to the range of variation in the sample available.

The coefficients of the quality variables in these regressions should be contrasted with the implicit valuations implied by the regressions of log quantity on log price and these variables and of transformed quantity on transformed price (Analyses B1 and B2) reported in Table 4. A likelihood-ratio test of Box–Cox transforms for quantity and price, respectively, with  $\theta = -0.78$  and  $\lambda = 0.06$  shows that the appropriate form of relationship is different from double log.

Regression results are summarized in Table 4. Table 5 gives the implied normalized implicit prices of each attribute obtained from the double log regres-

Table 5  
Comparison of selected results from Analyses A and B regressions, sugar included <sup>a</sup>

| Coefficients significant at 10% Level<br>in either regression | Normalized implicit prices from |                  |
|---------------------------------------------------------------|---------------------------------|------------------|
|                                                               | Analysis A                      | Analysis B       |
| Dependent variable                                            | $\log P$                        | $\log Q^*$       |
| <i>B</i>                                                      | -0.370<br>(4.51)                | 1.271<br>(2.91)  |
| <i>R</i>                                                      | -0.619<br>(2.66)                | 0.041<br>(0.07)  |
| <i>SOCKER</i>                                                 | -0.009<br>(1.26)                | -0.056<br>(2.87) |
| <i>ETAC</i>                                                   | 0.007<br>(4.91)                 | -0.001<br>(0.28) |
| <i>EXTR</i>                                                   | 0.008<br>(1.17)                 | 0.050<br>(2.69)  |
| <i>TOTSYR</i>                                                 | -0.001<br>(0.52)                | -0.022<br>(3.04) |
| <i>FN</i>                                                     | 0.005<br>(1.27)                 | -0.018<br>(1.73) |
| <i>BO</i>                                                     | 0.010<br>(2.09)                 | -0.010<br>(0.80) |
| <i>SY</i>                                                     | -0.007<br>(1.42)                | 0.025<br>(1.66)  |
| <i>ST</i>                                                     | 0.008<br>(1.89)                 | -0.004<br>(0.30) |
| <i>K</i>                                                      | 0.058<br>(1.97)                 | 0.065<br>(0.82)  |
| <i>BE</i>                                                     | 0.014<br>(2.38)                 | 0.029<br>(1.68)  |

<sup>a</sup> *t*-ratios in parentheses computed from asymptotic standard errors: asymptotic

$$\text{var} \left[ \frac{\hat{\beta}}{\hat{\gamma}} \right] = \frac{\hat{\beta}^2}{\hat{\gamma}^4} \text{var}(\hat{\gamma}) + \frac{1}{\hat{\gamma}^2} \text{var}(\hat{\beta}) - 2 \frac{\hat{\beta}}{\hat{\gamma}^3} \text{cov}(\hat{\gamma}, \hat{\beta}) > 0$$

, where  $\gamma$  is the coefficient of  $\log P$ , and  $\beta$  is the coefficient of a quality or other variable.

sion. These are obtained by dividing the coefficient of the  $i$ th nonprice variable by the absolute value of the coefficient of the price in the regression and normalizing by the price of the product variety. (Prices average 90–100 Skr.) These normalized implicit prices may be compared directly with the coefficients in the regressions reported in Table 3. Doing so reveals some remarkable differences and how misleading the confluent regression of transformed price on quality attributes can be. Similar comparisons are possible using the Box–Cox transformation reported in Table 3 and 4 but do not differ greatly from those reported in Table 5.

The effect of being imported in bulk is now positive. One should not conclude, however, that, adjusted for price, Swedish consumers prefer the cheaper bulk wines given other measurable quality attributes. Rather the positive coefficient of  $B$  is the result of the fact that bulk wines stock out less frequently than wines imported in bottles. Swedish consumers appear to prefer white wine and dryer and less acid wines. Extract (*EXTR*), which presumably consists largely of precipitated residual sugars, now has a positive but insignificant value, suggesting the presence of a more complicated interaction between sugar and acidity. Ethylacetate, the effect of which was too hard to interpret before, now drops to insignificance. Many of the remaining tasters' evaluations which were significant in the regressions with log price as dependent are now insignificant. Some such as color and color intensity which are now significantly negative. Tasters' overall evaluation remains a positive factor but is reduced to insignificance.

One of the most remarkable shifts is the change in the coefficient of ethylacetate (*ETAC*) from significantly positive to insignificantly negative. Ethylacetate content significantly affects the taste of wine: too little and the wine seems flat, some would say to lack 'character', too much and the wine tastes vinegary. The trick is to get the *ETAC* content high but not too high. Conversations with a well-known California wine-maker suggest the difficulties involved are reflected in higher prices for the high *ETAC*-content wines (presumably wines with too much *ETAC* are not imported into Sweden), so that the effect in the usual hedonic analysis is a supply-side phenomenon rather than a reflection of consumer preferences, *ceteris paribus*, given variety prices.

#### 4. Conclusions

Almost all hedonic analyses are based on regressions of unit variety prices on measures of quality characteristics (hedonic price functions) and almost all such analyses purport to infer implicit consumer valuations of these quality characteristics from the gradient of the hedonic price function. While this form of analysis may be justified if supplies of varieties are exogenously determined (polar case 1), the inferences may be seriously misleading if not. Another case (polar case 2) in which unit variety prices may be taken as exogenous is analyzed in this paper. The implied consumer implicit valuations are very different from those based on the

standard hedonic analysis. The general conclusion must be, I think, that the assumptions which underlie hedonic analysis along usual lines not only may not be justified in practice but make a very considerable difference to the results obtained.

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